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Research article

Mixed-methods research in the age of analytics, an exemplar leveraging sentiments from news articles to predict firm performance

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ABSTRACT

Investors and companies have always aspired to make informed investment decisions by using diverse information sources. With the explosion of information sources on the web and emergence of predictive analytics, many investors moved beyond traditional financial measures, as key predictors of firm performance, to textual content from analysts' reports. Empirical research suggests that these information sources complement each other by providing a clear picture of firm performance, but remains silent on the role of additional textual content that continues to emerge and reach more potential investors on the web. We build on this line of research to examine the effect of textual content from business journals in conjunction with summary measures on cumulative abnormal returns. We use sentiment analysis with machine learning and econometrics methods to examine content extracted from textual articles about S&P 500 index companies that are published in the Wall Street Journal (years 2013–2016). Textual analysis of business journals in conjunction with quantitative measures revealed direct and interaction effects on abnormal returns over time. We also tested for robustness by replicating the analysis with different variable operationalization and observe consistent patterns. Relative to positive sentiments, negative sentiments have more profound effects on cumulative abnormal returns. The effect of positive sentiments becomes weaker when past quantitative measures are high. As information sources continue to emerge on the web, this work makes key contributions to the practice of sentiment analysis in financial markets.

1. Introduction

> "You can get significant analytical lift when you combine information from the variety of sources available to you… Integrating and correlating across channels gives a more complete and useful insight into what is happening in a business" (Mighty Guides, 2016).

Recent market dynamics confirm the need for paying closer attention to diverse information sources about companies. For instance, share prices of the top-24 companies representing different industry sectors fell considerably from 45.6% to 14.6% after the news about the data breach in these companies hit the market (Comparitech, 2017). The breaches contributed to the spread of negative sentiments against the companies and ultimately, lowered their stock prices. Global equity markets dropped by $29 trillion in the 2016 crisis. During this global crisis, some companies recorded profits because they relied not only on past historical stock prices to project growth, but also on external factors such as news (Lee et al., 2017). Such an incident is not surprising given our knowledge of the importance of information sources that go beyond typical quantitative predictors of companies' performance (e.g., financial data), as suggested by earlier studies in the finance and accounting disciplines (Huang, Zang, & Zheng, 2014). Less is known, however, about the role of additional information sources that continue to evolve on the web.

As a key interdisciplinary field, information systems (IS) has diffused knowledge pertaining to information use on the web and analytical techniques with quantitative and qualitative or text data to offer insights about problems and opportunities in other disciplines (e.g., finance, marketing, accounting). Sentiment analysis, also known as opinion mining of text data, is the process of determining the emotional tone in a

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series of words to gain a deeper understanding of affective states that are present in text content (Pang & Lee, 2008). Several studies demonstrated the important role of sentiments resulting from technology adoption by organizations on their market value (Lee et al., 2017), stock returns from microblog sentiments (Deng, Huang, Sinha, & Zhao, 2018), tweets on stocks (Li, van Dalen, & van Rees, 2018), and online reviews on firm's profitability (Tang, Mehl, Eastlick, He, & Card, 2016). Such studies offer important insights based on sentiments, but leave two key areas that warrant further attention: (a) the role of sentiments from news articles in major news outlets, such as the Wall Street Journal (WSJ), whose scope cover diverse content (e.g., stories related to litigations and acquisitions) and (b) the interplay between sentiments from news articles and traditional quantitative financial variables for predicting firms' financial health.

Against this backdrop, we leverage theory-informed analytics and econometric techniques to move beyond addressing typical questions of "*what is hidden in the data*" to "*how hidden factors are inter-related*" or "*why is this so*" (Kar & Dwivedi, 2020, p. 2) in the context of financial measures and companies. We employ machine learning and econometric techniques to address the following research question: "*How do sentiments from business journals and traditional financial measures interact to predict firm performance?*" The scope of our investigation focuses on the direct and interaction effects among sentiments, key financial measures, and annual cumulative abnormal returns (i.e., returns generated by a given security or portfolio over a period of time that is different from the expected rate of return). We use Python and R programming approaches to extract the textual content and run the sentiment analysis, respectively. We extract the summary measures of S&P 500 index companies using Compustat and Eventus database. Textual content about firms (e.g., editorial opinions, market sentiments, litigations, competitive position, management policies, risks) plays a divisive role in the earnings of companies and thus represents important criteria that investors and stakeholders can use to assess financial performance.

Our work makes key contributions to research and practice. First, our study demonstrates the value of a synergistic framework of data and methods—offering the potential to extend the contemporary literature that uses computational methods to analyze textual information from business journals in financial markets (Hájek, 2018; Huang et al., 2014; Yu & Hatzivassiloglou, 2003). Specifically, using machine learning and econometrics methods with additional data sources complements recent work on harnessing the dual power of machine learning and econometrics methods in gaining richer insights (Adamopoulos, Ghose, & Todri, 2018; Wang, Li, & Singh, 2018). Second, our sample of S&P 500 index companies provides greater confidence for the generalizability of the predicted patterns. Third, using a multi-theoretic approach informs data extraction and enriches our understanding of the context of financial measures of companies. Finally, our results enlighten organizations about the importance of sentiments from business articles in explaining and predicting annual cumulative abnormal returns.

2. Theory and background

2.1. Information and firm performance

Relevant studies in finance and economics draw upon two key theories: information theory (Shannon, 1948) and adaptive market theory (Lo, 2004) to predict firm performance. Both theories inform empirical studies by emphasizing the value of information in reducing uncertainty about firm performance, but differ in the scope and role of information. Specifically, adaptive market theory (Lo, 2004) stresses that markets are not efficient and past summary measures cannot *fully* predict firm performance. Thus, adaptive market theory complements information theory (Shannon, 1948) by emphasizing the need for processing and utilizing diverse information sources to reduce uncertainty. For instance, studies in finance found that the impact of summary measures, such as earnings forecasts and target prices, on financial performance in the form of cash flows and revenues is significant (Asquith, Mikhail, & Au, 2005; Ertimur, Livnat, & Martikainen, 2003; Givoly et al., 2009). In a perfect information market, summary measures reflect all information pertaining to companies. In the era of digitization, however, information is easily accessible over the Internet (e.g., textual information published about the companies' stock prices, quarterly or yearly results, success, failures). Such diverse information plays an important role in the prediction of company growth (Altinkilic & Hansen, 2009). Hence, studies moved beyond quantitative measures of financial performance to incorporate textual information about companies. Choi et al. (2002) suggest that when investors use the web as an information channel, trading activity in corporate 401(k) plans doubles and portfolio turnover crosses over 50%. Tetlock (2007) emphasized the value of news stories for predicting earnings and stock returns. Heston and Sinha (2016) examined the role of news stories and sentiments on stock returns. Other studies developed a methodology to extract sentiments from the web (e.g., Das & Chen, 2007).

Recent IS research acknowledged the importance of information content in microblogs (Deng et al., 2018), tweets (Li et al., 2018), and reviews (Tang et al., 2016) for predicting firm performance in terms of stocks and firms' profitability (Kumar, Kar, & Ilavarasan, 2021). These studies, however, were either confined to analyzing a small subset of companies, analysts' reports, or limited time frame (Asquith et al., 2005; Huang et al., 2014; Huang, Lehavy, Zang, & Zheng, 2017). The analysts' reports mostly consist of raw data and company performance but may not capture market reactions in the event of litigations, success, etc. Market reactions play a key role in the future prospects of the company in the event of success or failure. Thus, embracing additional sources is important to gain insights on the market sentiments evolving about the company in reaction to (a) textual content published about the company and (b) summary measures announcements or past financial measures reflecting the growth of the company. This wide array of information represents an important resource for investors and companies. Investors will leverage this resource in their investment decisions and companies can take proactive approaches that are based on accurate predictions.

Overall, the existing body of knowledge has reached a consensus on the value of diverse information sources (El-Haj et al., 2019), but has not sufficiently examined approaches to leverage these sources in predicting firm performance. Consistent with Huang et al. (2014), for instance, Hájek (2018) examined financial indicators and qualitative data using analysts' reports to assess their direct effects on abnormal stock returns. Building on this line of research, we observe an opportunity to explore direct and interaction effects with different methods and additional information sources as suggested in the next section.

2.2. A synergistic framework of information sources and methods

Motivated by theory-based insights from prior work in (e.g., Hájek, 2018; Huang et al., 2014) and lack of studies that integrate sentiments from reputable business journals with summary financial measures, we leverage analytics to examine the interplay between quantitative financial measures and sentiments. Fig. 1 depicts our proposed framework for leveraging information sources and methods to explain and predict firm performance. Overarching frameworks provide researchers with systematic approaches to generate insights from data and guide other researchers who can continue building on research with analytics or machine learning (cf. Grover, Kar, & Davies, 2018; Jimenez-Marquez, Gonzalez-Carrasco, Lopez-Cuadrado, & Ruiz-Mezcua, 2019). Our framework depicts a sequence of phases that start with identifying relevant variables to build a model for explaining the phenomenon. In this phase, we highlight the complementary role of qualitative data and map it to an extraction method (Grover, Kar, & Dwivedi, 2020; Grover, Kar, & Ilavarasan, 2019; Kar, 2020). The second phase involves specifying and estimating an explanatory model to shed light on an understudied relationship. This phase is recommended prior to leveraging predictive analytics to enrich our understanding of firm performance or

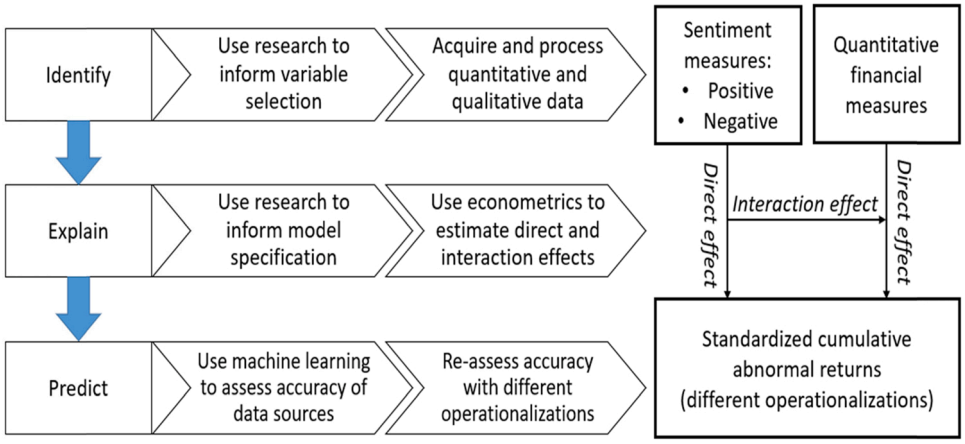


Fig. 1. A framework of information sources and methods.

the target variable using different data sources (Shmueli & Koppius, 2011). In the final phase, we predict firm performance to assess accuracy and errors using machine learning techniques. Such an assessment is critical when the target variable is already deployed in practice (Shmueli & Koppius, 2011).

Using theory-based insights (Kar & Dwivedi, 2020), we examine the role of news opinions about S&P 500 companies in conjunction with historical summary measures to shed light on their direct and interaction effects on firm performance in terms of the standardized cumulative abnormal return (SCAR). From scientific and practical perspectives, our predicted outcome is relevant as it (a) represents an important anomalous state and (b) helps in distinguishing between excellent investment opportunities/firms and investments to avoid. As noted in the upper right portion of the framework, we assess sentiment as a key dynamic that will shape the effects of traditional quantitative financial measures. Consistent with Huang et al. (2014), we suggest that sentiment from the textual content in conjunction with quantitative measures will provide a clearer picture of firm performance, as it captures evolution in the industry by reflecting change in management policies, industry regulations, litigations firms, and competitors' actions. Hence, with textual content, investors and stakeholders could predict financial performance effectively and make informed decisions. However, it is unclear whether positive/negative sentiments from business journals strengthen/weaken the relationship between quantitative financial measures and cumulative abnormal returns.

Emotive content present in the textual content can affect investors in different ways (Kumar et al., 2021). For instance, some could place more emphasis on the assertiveness, whereas others could get affected by the positiveness associated with the textual content, etc. In contrast, quantitative past summary measures provide only aggregated information that does not capture the present market sentiments about the company. For example, as soon as the news hit the market that Facebook compromised millions of users' data, stock price of Facebook took a nose dive (St. John, 2018). Investors became cautious in investing on Facebook stocks as data breach resulted into litigations against Facebook. Relying on past summary measures would have resulted into bullish market for Facebook but sentiments portrayed the bearish market for Facebook and saved many investors from losing money on the Facebook stocks. Thus, use of market sentiments and past summary measures will enrich the understanding of the investors about the projected company growth. Investors could adjust projections derived solely from past quantitative measures by incorporating the positiveness/negativity about the company present in the news.

Positive sentiments influence consumer behaviors (Adamopoulos et al., 2018), purchase decisions and reviews (Ahmad & Laroche, 2017), review ratings on hotel performance (Hu & Chen, 2016), and star ratings on bank's profitability (Tang et al., 2016). Negative sentiments about the company tend to generate unfavorable perceptions in investors' minds (Goode, Hoehle, Venkatesh, & Brown, 2017). Extant research has shown that negative sentiments lead to more uncertainty and ambiguity in the consumer mind. For instance, research suggests that consumers find negative reviews on service failure to be more helpful in ecommerce settings (Ahmad & Laroche, 2017). Similarly, Deng et al. (2018) note that negative sentiments in microblogs play a major role in stock returns. Hence, we hypothesize that *sentiments from business journals will shape the relationship between quantitative financial measures and cumulative abnormal returns*. In the following sections, we demonstrate how our approach to theory-based variable selection and analysis contributes meaningfully to our research objective (Kar & Dwivedi, 2020).

3. Methods

3.1. Quantitative and qualitative data acquisition

We adopted a structured approach to eliminate potential noise that may arise due to variety of data sources (Kar & Dwivedi, 2020). For instance, quantitative measures may be applied differently to operationalize constructs in predictive models. Qualitative data may suggest different signals depending on mechanisms for extracting and analyzing words across different disciplines. Hence, the first phase in our framework identifies relevant variables based on research and validated approaches for extracting data.

Accounting and finance research (e.g., Barth, Beaver, & Landsman, 1998) inform our variable selection to represent quantitative financial measures. Data on the variables listed in Table 1 was collected using Compustat and Eventus for the S&P 500 index companies for the year 2013–2016. To ensure objectivity in our measures, we computed the following variables using standard accounting formulas as per the generally accepted accounting principles (GAAP). Further, our data extraction approach eliminates potential noise when combining multiple sources of data to build and validate an inferential model (Kar & Dwivedi, 2020) as we explain in the sections that follow.

Table 1
Financial data variable list.

Variable	Description
Datadate	fiscal year end date
Fyear	fiscal year
Tic	company code/ ticker symbol
Conm	company name
Csho	common shares outstanding for fiscal year
Ib	income before extraordinary items for fiscal year
Revt	total revenues for a fiscal year
Teq	stockholders' equity
Xsga	selling, general and administrative expenses
Mkvalt	total market value (also used as a proxy for firm size)
Prcc_f	fiscal year closing price
Returns	change in fiscal year closing price as a percentage of previous fiscal year closing price
Price to earnings ratio	proportion of total market to the income before extraordinary items for given fiscal year
Market to book ratio	proportion of total market to the stockholders' equity for given fiscal year
sales growth	change in total revenues for fiscal year as percentage of previous fiscal year total revenues
SCAR	standardized cumulative abnormal returns

returns (t) = [prcc_f (t) – prcc_f (t-1)] / [prcc_f(t-1)]

price to earnings ratio (t) = mkvalt(t) / ib(t)

market to book ratio (t) = mkvalt(t) / teq(t)

sales growth (t) = [revt (t) – revt (t-1)] / [revt (t-1)]

We extracted textual data from the Wall Street Journal for the S&P 500 index companies for the year 2013–2016. We developed a web scraping Python programming code to extract about 2500 journal articles representing 500 companies. The code determines and scrapes the date the article was published, web URL of the article and the textual content present in the article. Fig. 2 shows a snippet of a data file generated after running the Python script for the company Intel.

From the textual data, sentiment analysis, also known as opinion mining, was used to determine the emotional tone in a series of words and gain an understanding of the affective states present in the content of the text. We wrote an R program to extract the sentiments from the textual content of the articles obtained from Wall Street Journal for the S&P 500 index companies. The affective states present in the content for the companies were determined by using the McDonald-Loughran financial dictionary, a widely used dictionary in Accounting and Finance. McDonald-Loughran dictionary has six categories of words namely: positive, negative, uncertainty, superfluous/modal, litigious, constraining. By basing our word extraction on common dictionaries, we also eliminate noise in potential signals from the data and increase confidence in our procedures (Kar & Dwivedi, 2020; Kushwaha, Kar, & Dwivedi, 2021). Our study used only positive and negative sentiments, as these sentiments are widely used in accounting studies to examine the impact of sentiments on financial performance of the company (Huang et al., 2014). The dictionary mostly consists of irregular verbs and syllables derived using syllabification algorithm.[1]

The R program generated sentiment flat file for yearly textual articles per company. The file consists of total word counts present in each fiscal year for respective company. The output also contains the word counts for each fiscal year file of a company. The word counts are represented for each Loughran defined classification categories of words namely, positive, negative, constraining, litigious, uncertainty and superfluous. Fig. 3 shows part of the sentiment file generated using R for the company Intel.

3.2. Approach to model specification and validation

We used positive and negative affective counts from the extracted sentiment data and ignored other affective wordcounts produced by the R program, as markets are more sensitive to positive and negative emotions (Huang et al., 2014). We standardized the positive and negative wordcount by dividing with total wordcounts. We also mean centered the variables to examine the interaction effects thereby removing multicollinearity. We controlled for prior SCAR and firm size. Larger firms (i.e., those with higher market capitalization of equity) or firms that have been historically doing well in terms of abnormal returns may perform better in the future. Hence, we eliminate such confounding effects by specifying prior SCAR and firm size as control variables. Based on relevant accounting and finance studies, we used PE_ratio, sales growth and MB_ratio as quantitative measures (Asquith et al., 2005; Huang et al., 2014, 2017).

We employed generalized estimating equations (GEE) to specify a model to explain SCAR. As illustrated in Fig. 1, this specified model will become the basis for predicting SCAR using machine learning algorithms (i.e., the third phase in our framework of information sources and methods). For the econometric analysis, our model specification and validation approach is appropriate for our longitudinal dataset, as we are interested in estimating the average response for cumulative abnormal returns over the population rather than for a company. Further, the presence of autocorrelation in the data deems linear regression inappropriate. GEE takes into account autocorrelation and incorporates a time variable to control for this issue. For our analysis with sentiment data, we also used GEE (see Ballinger, 2004 for a detailed discussion of this technique). We strengthen our model specification and validation approach by replicating the underlying procedures (Kar & Dwivedi, 2020) to predict abnormal returns with an alternative operationalization or binary measurement of SCAR. As for predicting SCAR using machine learning algorithms in the third phase, we first chose algorithms that are appropriate for the measurement of the target/dependent variable or SCAR (Choudhury, Allen, & Endres, 2021). For instance, with a non-binary specification of SCAR, we adopted linear regression algorithms (e.g., boosted decision tree regression, neural network regression). With a binary specification of SCAR for robustness testing, we adopted classification algorithms (e.g., two-class Bayes point machine, two-class logistic regression, two-class boosted decision tree). Hence, our goal with machine learning was to obtain predictive accuracy of the explanatory model using different data sources rather than discover new patterns in predictors (Choudhury et al., 2021) that led to SCAR or to re-specify key relationships that have been typically examined in the literature.

4. Results

4.1. Econometric techniques

As discussed in Section 2.2., the second phase in our framework focuses on explaining the specified relationship that we developed based on research. From an analytical perspective, we use econometric techniques to assess the sign and significance of the direct and interaction effects—an approach that is recommended prior to applying machine learning techniques (Shmueli & Koppius, 2011). We ran the GEE model using SCAR as a continuous variable and later, binary variable as robustness testing (Kar & Dwivedi, 2020). Table 2 shows the GEE results with different correlation matrix structures (Ballinger, 2004). The results suggest that prior year SCAR has a significant negative effect on cumulative abnormal returns. Firm size and sales growth have a

[1] Details about the words included in each category could be found from the link - https://sraf.nd.edu/textual-analysis/resources/#LM%20Sentiment%20Word%20Lists. ***Positive***: positive feelings attributed such as abundance, growth, profit, achievement, etc. ***Negative***: Negative states such as losses, bankruptcy, decline, etc.

link	date	text
https://www.wsj.com/articles/apple-and-samsung-mi	12/29/2016 18:19	in the continuing smartphone battle between apple inc aapl 0 47 and samsung electronics co the week leading up to christmas didn t provi
https://www.wsj.com/articles/apple-ebay-and-gold-v	12/29/2016 11:38	predicting the future is typically a fool s game just ask any wall street strategist about their year end price targets and forecasting the reacti
https://www.wsj.com/articles/luxembourg-tightens-t	12/27/2016 9:02	brussels luxembourg published tightened tax rules on tuesday bending to pressure from the european union to crack down on corporate ta
https://www.wsj.com/articles/surging-buybacks-say-s	12/26/2016 13:18	corporate stock repurchases are on the upswing once again wrong footing skeptics who predicted 2016 would mark the beginning of the enc
https://www.wsj.com/articles/nokia-sues-apple-alleg	12/21/2016 18:26	apple inc and nokia corp filed competing lawsuits over intellectual property used in the iphone and other apple products the suits filed this
https://www.wsj.com/articles/japan-display-to-get-u	12/21/2016 7:46	apple inc aapl 0 47 supplier japan display inc 6740 0 50 on wednesday said it would receive 75 billion 636 million in aid from its largest sh
https://www.wsj.com/articles/apple-is-discussing-ma	12/20/2016 13:25	new delhi apple inc aapl 0 47 is discussing with the indian government the possibility of manufacturing its products in the country accordir
https://www.wsj.com/articles/ireland-says-eu-overst	12/19/2016 16:06	ireland said the european union overstepped its authority and misinterpreted irish law when it ordered the country in august to recoup 13 b

Fig. 2. Textual file for Intel generated from Python.

```
[1] 2671
# A tibble: 1 x 6
  constraining litigious negative positive uncertainty        file
         <dbl>     <dbl>    <dbl>    <dbl>       <dbl>       <chr>
1            1        12       48       12          25 FB_2013.csv
[1] 60234
# A tibble: 1 x 7
  constraining litigious negative positive superfluous uncertainty        file
         <dbl>     <dbl>    <dbl>    <dbl>       <dbl>       <dbl>       <chr>
1          104       438     1028      452           5         440 FB_2014.csv
[1] 71120
# A tibble: 1 x 7
  constraining litigious negative positive superfluous uncertainty        file
         <dbl>     <dbl>    <dbl>    <dbl>       <dbl>       <dbl>       <chr>
1          118       631     1236      483           2         472 FB_2015.csv
[1] 87780
# A tibble: 1 x 7
  constraining litigious negative positive superfluous uncertainty        file
         <dbl>     <dbl>    <dbl>    <dbl>       <dbl>       <dbl>       <chr>
1          135       489     1544      652           5         654 FB_2016.csv
```

Fig. 3. Part of the sentiment file generated using R for Intel.

significant positive effect on abnormal return. Consistent with Huang et al. (2014), positive sentiments have a significant positive effect on cumulative abnormal returns, whereas negative sentiments have a significant negative effect on cumulative abnormal returns. Interaction of positive sentiments and sales growth has a significant negative effect on abnormal returns. Such a pattern suggests that positive sentiments show a diminishing effect at higher values of sales growth. Interaction of positive sentiments and MB_ratio has a significant positive effect on abnormal returns. Such a pattern suggests that the magnitude of the positive sentiments effect becomes stronger at higher values of MB_ratio. Interaction of negative sentiments and sales growth has a significant negative effect on cumulative abnormal returns. Hence, the magnitude of the negative sentiments effect becomes stronger at higher values of sales growth. Finally, interaction of negative sentiments and MB_ratio has a significant negative effect on cumulative abnormal returns. Such a pattern suggests that the magnitude of the negative sentiments effect becomes strong at higher values of MB_ratio.

Table 3 shows estimation results from fixed effect and random effect models. The results suggest that prior year SCAR has a significant negative effect on cumulative abnormal returns. Firm size and sales growth have positive significant effects on abnormal return. Positive sentiments have a significant positive effect on cumulative abnormal returns. Negative sentiments have a significant negative effect on cumulative abnormal returns. Interaction of positive sentiments and sales growth has significant negative impact on abnormal returns. Such a pattern suggests that the magnitude of the positive sentiments effect becomes weaker at higher values of sales growth. Interaction of positive sentiments and MB_ratio have significant positive effects on abnormal returns. Hence, the magnitude of the positive sentiments effect becomes stronger at higher values of MB_ratio. Interaction of negative sentiments and sales growth has a significant negative effect on cumulative abnormal returns. Such a pattern suggests that the magnitude of negative sentiments effect becomes stronger at higher values of sales growth. Finally, interaction of negative sentiments and MB_ratio has a significant negative effect on cumulative abnormal returns. Hence, the magnitude of negative sentiments effect becomes stronger at higher values of MB_ratio. The results support our hypothesis in that there is a joint effect between sentiments from business journals and financial indicators, but reveal interesting dynamics about the differential effects of positive and negative sentiments.

We also studied the upward and downward trend of cumulative abnormal returns. We dichotomized the cumulative abnormal returns as high if the value was positive and low if the value was negative to further validate our model with a modified operationalization (Kar & Dwivedi, 2020). Table 4 shows the results from both fixed effect and random effect models in the classification of cumulative abnormal returns. The results suggest that prior year SCAR has a significant negative effect on cumulative abnormal returns. Firm size and sales growth have

Table 2
GEE results predicting SCAR.

Model: Correlation Matrix:	**1** Independent	**2** AR (1)	**3** Exchangeable
Prior_SCAR	-0.065** (0.024)	-0.065** (0.024)	-0.064** (0.024)
Size	0.355*** (0.063)	0.355*** (0.063)	0.355*** (0.063)
Positive	13.665* (5.711)	13.665* (5.711)	13.654* (5.710)
Negative	-11.119*** (2.974)	-11.119*** (2.974)	-11.117*** (2.974)
PE_ratio	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Sales_growth	1.225*** (0.285)	1.225*** (0.285)	1.225*** (0.285)
MB_ratio	0.001 (0.002)	0.001 (0.002)	0.001 (0.002)
PE_ratio X Positive	0.105 (0.093)	0.105 (0.093)	0.106 (0.093)
Sales_growth X Positive	-102.094** (39.289)	-102.094** (39.289)	-102.135** (39.286)
MB_ratio X Positive	0.027* (0.093)	0.027* (0.093)	0.027* (0.093)
PE_ratio X Negative	-0.044** (0.037)	-0.044** (0.037)	-0.044** (0.037)
Sales_growth X Negative	-30.020* (19.796)	-30.020* (19.796)	-30.026* (19.794)
MB_ratio X Negative	-0.052** (0.081)	-0.052** (0.081)	-0.052** (0.081)
Goodness of Fit:			
QIC	1442.801	1442.801	1442.802

Notes: $^{+}p < 0.1$;
* $p < 0.05$;
** $p < 0.01$;
*** $p < 0.001$.

significant positive effects on cumulative abnormal returns. Negative sentiments have a significant negative effect on cumulative abnormal returns. Interaction of positive sentiments and sales growth has significant negative impact on abnormal returns. Such a pattern suggests that the magnitude of the positive sentiments effect becomes weaker at higher values of sales growth. Interaction of positive sentiments and MB_ratio has a significant positive impact on abnormal returns. Hence, the magnitude of the positive sentiments effect becomes stronger at higher values of MB_ratio. Interaction of negative sentiments and sales growth have significant negative effects on cumulative abnormal returns. Such a pattern suggests that the magnitude of the negative sentiments effect becomes stronger at higher values of sales growth. Finally, interaction of negative sentiments and MB_ratio has a significant negative effect on cumulative abnormal returns. Hence, the magnitude of the negative sentiments effect becomes stronger at higher values of MB_ratio. For additional insights on the patterns, please refer to the interaction plots in the appendix.

4.2. Machine learning techniques

The third phase in our framework focuses on using machine learning techniques to predict firm performance and assess accuracy and errors with the specified relationship and data sources. We used several machine learning algorithms to test the accuracy of models using quantitative summary data and both quantitative and text data. Studies that apply machine learning in management disciplines typically use different specifications of decision trees, random forests, and neural networks (see Choudhury et al., 2021). In general, researchers and data scientists do not know a priori the best machine learning algorithm that fits their inquiry or situation (Choudhury et al., 2021). Given our objective of exploring the synergy among different data sources and methods, with special emphasis on prediction accuracy, we adopted two basic rules for deploying machine learning algorithms. First, we chose machine learning algorithms that fit the specification of SCAR (i.e., classification algorithms for the binary specification and regression-based algorithms for the non-binary specification). Decision

Table 3
Results from fixed effect and random effect models predicting SCAR.

	Fixed Effect	**Random Effect**
Prior_SCAR	-0.492*** (0.027)	-0.062* (0.026)
Size	5.493*** (0.319)	0.366*** (0.064)
Positive	16.621** (6.193)	11.791* (5.238)
Negative	-1.821 (3.365)	-11.544*** (2.779)
PE_ratio	-0.000 (0.000)	0.000 (0.000)
Sales_growth	0.409* (0.187)	0.956*** (0.151)
MB_ratio	0.001 (0.001)	0.001 (0.001)
PE_ratio X Positive	0.112 (0.098)	0.102 (0.093)
Sales_growth X Positive	-15.313* (40.252)	-99.499** (36.534)
MB_ratio X Positive	0.070* (0.183)	0.027* (0.138)
PE_ratio X Negative	-0.002 (0.045)	-0.043 (0.044)
Sales_growth X Negative	-12.527** (16.980)	-27.957^{+} (15.739)
MB_ratio X Negative	-0.044** (0.099)	-0.051** (0.083)
Goodness of Fit:	**F-statistic**	**Wald chi^2**
	42.96	107.25

Notes: $^{+}p < 0.1$;
* $p < 0.05$;
** $p < 0.01$;
*** $p < 0.001$. Standard errors are in parentheses.

Table 4
Results from fixed effect and random effect models classifying SCAR.

	Fixed Effect	**Random Effect**
Prior_SCAR	-1.058*** (0.108)	-0.127* (0.056)
Size	12.747*** (1.323)	0.605*** (0.143)
Positive	14.745*** (20.332)	16.159*** (11.665)
Negative	-8.893* (10.558)	-16.652** (5.996)
PE_ratio	0.002 (0.002)	0.001 (0.001)
Sales_growth	0.670** (0.577)	1.230** (0.356)
MB_ratio	0.001 (0.003)	0.001 (0.002)
PE_ratio X Positive	-0.648 (0.511)	-0.055 (0.281)
Sales_growth X Positive	-107.300* (133.088)	-122.894* (83.614)
MB_ratio X Positive	0.733** (0.765)	0.245** (0.311)
PE_ratio X Negative	-0.098 (0.126)	-0.091 (0.102)
Sales_growth X Negative	-13.936* (59.502)	-70.471^{+} (37.477)
MB_ratio X Negative	-0.150* (0.304)	-0.101* (0.187)
Goodness of Fit:	**LR chi^2**	**Wald chi^2**
	246.27	46.96

Notes: $^{+}p < 0.1$;
* $p < 0.05$;
** $p < 0.01$;
*** $p < 0.001$. Standard errors are in parentheses.

Table 5
Cumulative abnormal returns prediction - quantitative data only.

Algorithm	Mean Abs Error	Root Mean Squared Error	Relative Abs Error	Relative Squared Error	Coefficient of Determination
Bayesian Linear Regression	0.538	3.425	2.440	128.503	-127.503
Boosted Decision Tree Regression	0.255	0.340	1.156	1.270	-0.270
Decision Forest Regression	0.237	0.321	1.073	1.131	-0.131
Linear Regression	0.430	2.358	1.948	60.937	-59.937
Neural Network Regression	0.224	0.306	1.014	1.027	-0.027

Table 6
Cumulative abnormal returns prediction – quantitative data and text metrics.

Algorithm	Mean Abs Error	Root Mean Squared Error	Relative Abs Error	Relative Squared Error	Coefficient of Determination
Bayesian Linear Regression	0.388	2.770	1.590	56.259	-55.259
Boosted Decision Tree Regression	0.250	0.386	1.023	1.091	-0.091
Decision Forest Regression	0.217	0.341	0.890	0.850	0.149
Linear Regression	0.372	2.490	1.524	45.446	-44.446
Neural Network Regression	0.265	0.450	1.086	0.484	-0.484

Table 7
Cumulative abnormal returns classification – quantitative data only.

Algorithm	Accuracy	AUC
Two-Class Bayes Point Machine	70.7%	62.9%
Two-Class Averaged Perceptron	70.0%	58.2%
Two-Class Boosted Decision Tree	67.9%	54.0%
Two-Class Decision Forest	68.6%	48.7%
Two-Class Decision Jungle	68.8%	52.2%
Two-Class Locally-Deep Support Vector Machine	69.3%	54.1%
Two-Class Logistic Regression	72.9%	38.8%
Two-Class Neural Network	71.4%	51.5%
Two-Class Support Vector Machine	72.1%	49.5%

Table 8
Cumulative abnormal returns classification – quantitative data and text metrics.

Algorithm	Accuracy	AUC
Two-Class Bayes Point Machine	84.0%	85.7%
Two-Class Averaged Perceptron	78.4%	79.0%
Two-Class Boosted Decision Tree	99.7%	99.9%
Two-Class Decision Forest	83.9%	86.0%
Two-Class Decision Jungle	82.8%	84.7%
Two-Class Locally-Deep Support Vector Machine	80.6%	81.8%
Two-Class Logistic Regression	82.8%	85.2%
Two-Class Neural Network	79.7%	74.0%
Two-Class Support Vector Machine	78.2%	78.7%

trees, random forests, and neural networks can be adjusted to work with binary and non-binary specifications of the dependent variable (Choudhury et al., 2021). Second, we chose algorithms that offer medium to high capacity—algorithm's capacity to achieve high predictive performance (Choudhury et al., 2021). For classification and regression-based prediction, decision trees, random forests, and neural networks also satisfy the objective of prediction performance (Choudhury et al., 2021).

We used 60% of the data for training and 40% for prediction or validation. Our approach echoes recent calls to eliminate potential bias by leveraging diverse data sources from which theory-based variables can be extracted (Kar & Dwivedi, 2020). We found that accuracy of the models was quite high and errors were low when both quantitative and textual data was used. Table 5 shows the results of machine learning algorithms using only quantitative data. Table 6 shows the results of machine learning algorithms using both quantitative and text data in predicting cumulative abnormal returns. On comparing Tables 5 and 6, we notice that textual data in conjunction with quantitative data results in better accuracy and low errors. We also used machine learning techniques to compare the accuracy of the classification of cumulative abnormal returns. We notice from Tables 7 and 8 that text data along with quantitative data provide better accuracy as compared to quantitative data used alone in the classification of cumulative abnormal returns. For instance, two-class bayes point machine provided 84% accuracy when both quantitative and text data was used as compared to the accuracy of 70.7% when only quantitative data was used.

5. Discussion and implications

As diverse information sources continue to explode on the web, investors and companies will find systematic approaches that embrace these sources indispensable for investment decisions. Building on existing research, we propose a framework of synergistic data sources and methods to move beyond addressing typical questions of "*what is hidden in the data*" to "*how hidden factors are inter-related*" or "*why is this so*" (Kar & Dwivedi, 2020, p. 2) in the context of financial measures and companies. We employ both machine learning and econometric techniques to examine the effects of sentiments from news articles and financial measures on annual cumulative abnormal returns. Our findings suggest that sentiments introduce important dynamics that shape the effects of quantitative financial measures on annual cumulative abnormal returns. The findings were consistent across different model estimations—suggesting positive effects of positive sentiments and negative effects of negative sentiments on cumulative abnormal returns. We also found that high sales growth and MB_ratio generate high cumulative abnormal returns but the effect diminishes when the positive sentiments in the market are high. On the contrary, higher negative sentiments subsides the influence of higher sales growth and MB_ratio on cumulative abnormal returns. We extended the econometric analysis

using machine learning techniques and found that textual data in conjunction with quantitative measures provide higher prediction accuracy of cumulative abnormal returns.

5.1. Theoretical implications and expected contributions

The main objective of the proposed framework is to illustrate the value of diverse information sources and methods. Although our work presents an exemplar, our findings introduce a new lens for potential theory development in two ways. First, our framework offers guidance with problem formulation using analytics. Specifically, we propose a realistic view of firm performance by examining interactions of predictors that have been typically examined as direct effects. For instance, Hájek (2018) used financial indicators and qualitative data to examine direct effects on abnormal stock returns but did not consider interaction effects tied to new data sources. Hence, our findings complement the finance and accounting literature on predictive analytics in the context of financial performance of companies. Prior literature has studied the role of online word-of-mouth and sentiments, mainly using analysts' reports, in predicting key performance measures of companies, but remained silent on the role of news articles on cumulative abnormal returns of the company. Our study highlighted the importance of diverse information by using information theory and market adaptive theory in the new context (see Johns, 2006).

Second, our framework offers guidance with leveraging mixed-methods and setting a threshold for sample size and accuracy and error rates. For instance, a sample of S&P 500 index companies provides greater confidence for the generalizability of the results. With higher accuracy and lower error rates, our work complements prior literature on word-of-mouth by rigorously demonstrating that sentiments from business journals play an important role in the prediction of financial measures of a company. Further, we developed a web-scraping program in Python to extract rich text for each company on a day to day basis. Our dataset comprises diverse information content about the examined companies. Finally, our use of machine learning and econometric techniques opens avenues for future researchers to deploy similar effective approaches and unearth insights from textual information.

5.2. Limitations and future research

Our results indicate that sentiments play a major part in predicting financial performance of the companies. We used longitudinal data for S&P 500 index companies from news articles published in the Wall Street Journal. However, future research could build on our study in three ways. First, although the Wall Street Journal is a reputable source with high coverage, future research could study news articles published in other business journals. Second, future research could reassess the interaction effect in our model (i.e., identify an additional moderator) using more sources and approaches to sentiment analysis, such as social media and topic modeling in general (Kar, 2020) or content pertaining to corporate social responsibility in particular (Grover et al., 2019). Finally, we studied sentiments from news articles published in 2013–2016. Future research could build on our approach to extract data over longer and other periods of time, as longer and/or different manifestations of time may reveal additional insights on the phenomenon of interest (Venkatesh, Maruping, & Brown, 2006; Venkatesh, Sykes, Aljafari, & Poole, 2021).

5.3. Implications to practice

Our findings offer important practical guidance, especially with our focus on a key outcome for distinguishing between excellent investment opportunities and firms and/or investments and sentiments from news articles. Investors should not rely solely on past summary measures given their interaction effects with sentiments from news articles. Given that sentiment analysis is widely used in practice (Kumar et al., 2021), managers could adopt our approach and consider employing sentiments in their analytical models when forecasting the growth of their companies. Although positive sentiments could foster growth as management can engage in expansion of the company, negative sentiments caution the management and provide an opportunity to take corrective actions before the company starts to fall apart. Also, our study highlights that positive sentiments do not play a major role when the quantitative performance of the companies is high and the effect of positive sentiments diminishes over a period of time. On the contrary, negative sentiments play an important role and subsides the influence of past quantitative measures in the prediction of cumulative abnormal returns.

6. Conclusion

Diverse information sources on the web and the emergence of predictive analytics emphasize the need to move beyond traditional financial measures as key predictors of firm performance. Although empirical research suggests that information sources from particular sources and financial measures complement each other by providing a clear picture of firm performance, less is known about the role of additional textual content that continues to emerge and reach more potential investors on the web. Using additional textual content, we demonstrate a valuable synergy among sentiment analysis, machine learning, and econometrics methods to predict abnormal returns over time. We examine content extracted from the textual articles for S&P 500 index companies published in the Wall Street Journal (years 2013–2016). We find that textual analysis of business journals in conjunction with quantitative measures introduce direct and interaction effects, as sentiments shape the effects of quantitative measures on cumulative abnormal returns. Our theory-based approach offers important practical insights about the role of sentiment in conjunction with past quantitative summary measures.

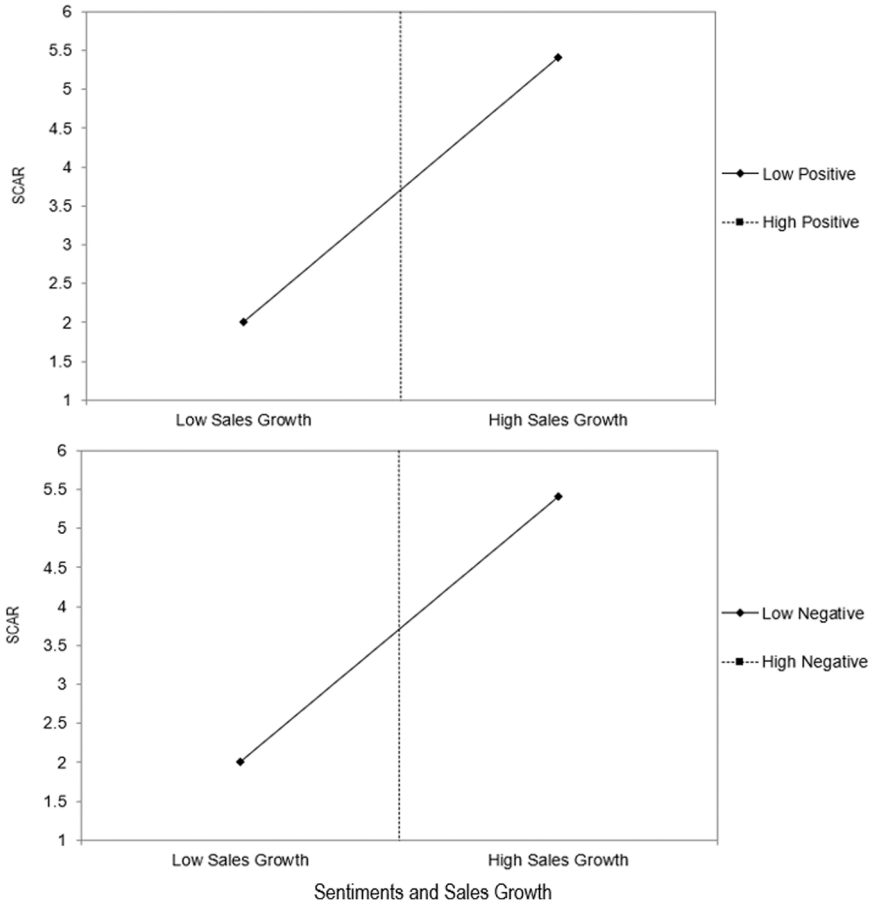
CRediT authorship contribution statement

Conceptualization: Raman, Data curation: Raman with support from graduate assistant, Formal analysis: Raman with support from graduate assistant & Venkatesh, Methodology: Raman & Venkatesh, Roles/Writing - original draft & revision: Raman & Aljafari, Writing – review & editing: Venkatesh & Richardson.

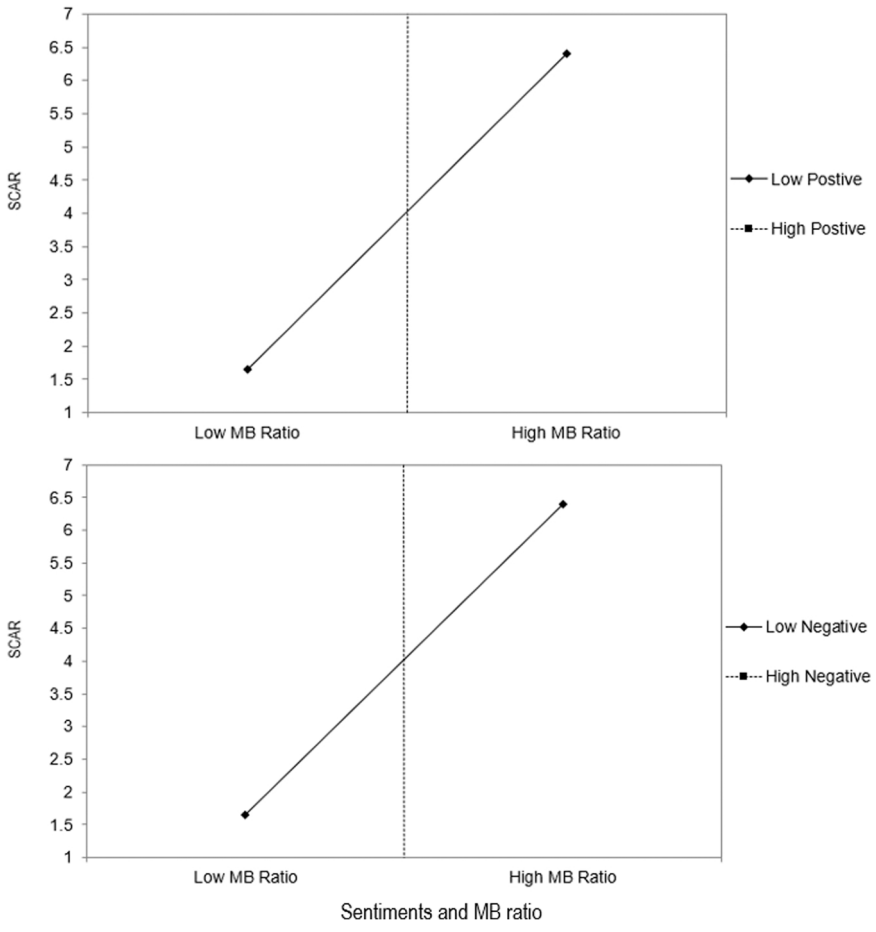
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Appendix. Interaction plots



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